

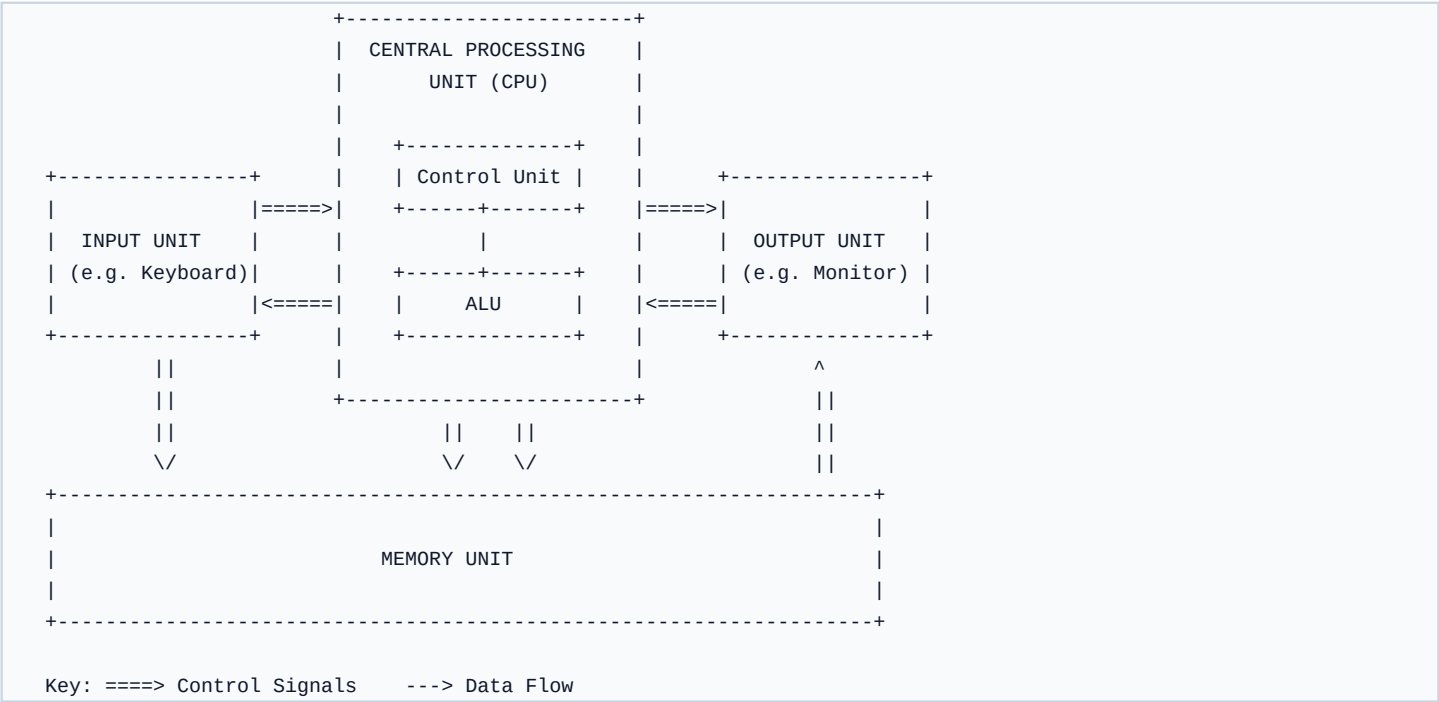
Block-Level Description of Functional Units (Self-Learning)

Q. Draw a block-level diagram of a computer system. Describe how the functional units are interconnected and how data and control signals flow between them.

[DEF] Definition to Remember

A Block-Level Description illustrates the physical grouping of a computer's functional units (Input, Output, Memory, CPU) and how they communicate. They do not operate in isolation; they are interconnected via a Bus System that transports data, memory addresses, and control signals.

1. Block Diagram of Functional Units



2. The Bus System (Interconnection)

Units communicate over a Bus System:

3. System Flow

1. Input: Control Unit sends a *Read* signal; Input data travels over the *Data Bus* to Memory.

1. Functional units are interconnected using a Bus System.
2. Block Diagram shows CPU (CU + ALU) connected to Memory, Input, and Output units.
3. Data Bus: Bidirectional; carries actual data.
4. Address Bus: Unidirectional; carries addresses from CPU to Memory/IO.
5. Control Bus: Carries synchronization signals (Read/Write) from the Control Unit.
6. Control Unit directs traffic via control signals (====>).

7. Data (--->) always flows through the Memory Unit before going to the CPU or Output.

[RECALL] Quick Recall

Block Diagram -> CPU, Memory, I/O -> Interconnected via Buses -> Data Bus (2-way, data) -> Address Bus (1-way, location) -> Control Bus (Signals: Read/Write)